FEDERAL SPECIFICATION

GRADUATE, LIQUID, LABORATORY; PLASTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements of plastic laboratory graduates.

1.2 Classification.

1.2.1 *Types and sizes.* Plastic graduates shall be of the following types and sizes, as specified (see 6.2):

Type I—Polyethylene.

Size 1—250 milliliters (ml.); 8 ounces (oz.).

Size 2—500 ml.; 16 oz.

Type II—Polypropylene.

Size 1—250 ml.; 8 oz.

Size 2—500 ml.; 16 oz.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-590—Plastic Compounds, Molding and Extrusion, Polyethylene.

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).

PPP-B-566—Boxes, Folding, Paperboard.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676—Boxes, Set-Up Paperboard.

PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Standard No. 406—Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

FSC 6640

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Military Specification:

MIL-L-10547—Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Type I.* The polyethylene used for type I graduates shall conform to type I, grade 1 of L-P-590.

3.1.2 *Type II.* The polypropylene used for type II graduates shall conform to the requirements of table I.

TABLE I.—*Polypropylene requirements*

Requirement	Limit	Test method (Fed. Test Method Std. No. 406)
Specific gravity	0.897 to 0.903	5011
Tensile strength, minimum psi	4900	1011
Water absorption, maximum	0.01	7031
Brittleness temperature, °C.	-15	2051

3.2 Design.

3.2.1 *Pouring lip.* Graduates shall have a well defined pouring lip and shall pour evenly when tested as specified in 4.6.1.

3.2.2 *Shape.* Graduates shall be conical, without handle, and molded, having a flat bottom perpendicular to the vertical axis of the graduate and tapered sides extending from the bottom away from the vertical axis to the top edge of the graduate.

3.2.3 *Graduation markings.*

3.2.3.1 *General.* Each graduate shall have clear etched or molded graduation markings, perpendicular to the vertical axis of the graduate and parallel to each other. Each graduate shall be graduated in both the metric and apothecary scales. Each main graduation shall be marked to show its value; intermediate graduations shall not be marked. Value figures shall be uniformly positioned, in Arabic numerals, at the end of the graduations to which they refer.

3.2.3.2 *Markings.* The markings, as specified in table II, shall appear on the outside of the graduate when viewed with the pour lip on the left. The markings shall consist of a center line extending from the capacity graduation, perpendicularly, to the initial or lowest value graduation. The metric scale graduations shall extend from (but not through) this center line to one side; the apothecary scale graduations shall extend from (but not through) this center line to the opposite side (of the metric scale). The width of the graduations shall not exceed 0.04 inch (1.0 millimeters). For size 1 graduates, the main graduations and intermediate graduations, respectively, shall have a maximum length of 11/32 and 1/4 inch. For size 2 graduates, the main graduations and intermediate graduations, respectively, shall have a maximum length of 7/16 and 5/16 inch.

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TABLE II.—*Marking details for graduates*

Capacity	Intermediate graduations (not numbered)	Main graduations (numbered)
250 ml.	10, 20, 30, 40, 60, 70, 80, 90, 110, 120, 130, 140, 160, 170, 180, 190, 210, 220, 230, 240 (ml.)	50, 100, 150, 200, 250 (ml.)
8 oz.	½, 1½, 2½, 3½, 4½, 5½, 6½, 7½ (oz.)	1, 2, 3, 4, 5, 6, 7, 8 (oz.)
500 ml.	150. 250, 350, 450 (ml.)	50, 100, 200, 300, 400, 500 (ml.)
16 oz.	3, 5. 7, 9, 11, 13, 15 (oz.)	1, 2, 4, 6, 8, 10, 12, 14, 16 (oz.)

3.2.3.3 *Tolerances.* All graduation markings shall be accurate to within plus or minus 3.0 percent of the indicated volume as specified in 4.6.2.

3.3 **Color.** The graduate shall be the natural color of the plastic material. Type I graduates shall be opaque. Type II graduates shall be translucent.

3.4 **Identification marking.** Graduates shall be permanently and legibly marked with the name or trademark of the manufacturer.

3.5 **Workmanship.** Graduates shall be uniform in thickness, color, texture, and finish and shall be free from porosity, warpage, checks, cracks, chipped edges, blisters, scratches, and other defects that impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the plastic graduates of the same type and size, offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions.

4.4 **Sampling.**

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* One graduate per each 100 graduates or fraction thereof in each lot shall be taken at random as a sample. In no case shall fewer than 3 graduates be taken as a sample. Each graduate in the sample shall be subjected to all tests specified in 4.6.

4.5 **Examination.** Samples of graduates

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and level A and level B preparation for delivery requirements shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Graduates.*

Categories	*Defects*
Critical:	None defined
Major:	*AQL 4.0 percent defective*
101	Base not perpendicular (visually) to vertical axis
102	Graduation markings or values not as specified
103	Graduate not uniform in thickness, color, texture, or finish
104	Graduate damaged (porous, warped, checks, cracks, chips, blisters, or scratches)
105	Graduation markings or identification markings illegible, incorrect, or incomplete

4.5.1.2 *Preparation for delivery (section 5).*

Categories	*Defects*
Critical:	None defined
Major:	*AQL 2.5 percent defective*
101	Unit or shipping container not as specified
102	Unlike graduates packed per shipping container
103	Unit or shipping container damaged
104	Blocking or cushioning not as specified
105	Marking of unit or shipping container illegible, incorrect, or incomplete

4.6 **Tests.** Distilled water shall be used for all tests.

4.6.1 *Pour test.* Fill the graduate to the indicated capacity with water. Pour approximately one-fourth of the water from the graduate at such a rate that the water flows in a continuous stream and falls free from the lip of the graduate. If the stream of water clings to the outside of the graduate and flows off the bottom, it shall be considered cause for rejection.

4.6.2 *Tolerances.* The accuracy of the graduation markings shall be determined at 10 percent, 50 percent, and 100 percent of capacity by filling the tared (to the nearest 0.1 gram) graduate to the desired marking for test with water at room temperature, and reweighing the graduate and water to the nearest 0.1 gram, noting the temperature of the water.

$$A = \frac{100\ (W_2 - W_1)\ f_t}{V} - 100$$

where:

A = variation from 100 percent accuracy of marking,

W_2 = weight, in grams, of the graduate and water,

W_1 = tare weight of graduate, in grams,

f_t = reciprocal of relative density of water at temperature t (milliliters/gram),

t = temperature of water, and

V = volume at test marking, in milliliters.

4.7 **Acceptance/rejection criteria.** If any graduate fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurements, the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

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5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Each graduate shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636, type I, class 1; or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to prevent shifting of contents. Box closure shall be secured with tape conforming to UU-T-111.

5.1.2 *Level C.* Graduates shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Graduates shall be packed in a box conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. The container shall be provided with a liner conforming to MIL-L-10547 which shall be closed and sealed in accordance with the appendix thereto. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 *Level B.* Graduates shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1 or PPP-B-601, domestic type. Case liners shall not be required.

5.2.3 *Level C.* Graduates shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Plastic graduates are intended for use where the volumetric requirements for accuracy are not as exacting as for precise scientific laboratory work and where it is desirable to avoid breakage of glassware.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and size required.

(*c*) Level of packaging and packing required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Transportation description.** Transportation description applicable to this item is:

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Plastic articles not otherwise indexed by name. (Specific weight per cubic foot).

Carload minimum weights—10,000, 20,000, and 30,000 pounds, subject to Rule 34, Uniform Freight Classification.

Plastic articles, not otherwise indexed. (Specify density).

Truckload minimum weights—10,000, 8,600, 7,100; 16,000, 13,700, 11,400; 21,000; 18,000, 15,000; 30,000, 25,700, and 21,400 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—MU

Navy—Md

Air Force—DAY

OTHER INTERESTS:

Army—O E M

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